RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Thursday, 8 October 2026, 11:13 AM
State	Finished
Completed on	Friday, 9 October 2026, 3:34 PM
Time taken	1 day 4 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char s1[1000], s2[1000];
    int dp[1001][1001];
    int m, n, i, j;

    scanf("%s", s1);
    scanf("%s", s2);

    m = strlen(s1);
    n = strlen(s2);

    for (i = 0; i <= m; i++) {
        for (j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = 1 + dp[i - 1][j - 1];
            } else {
                dp[i][j] = (dp[i - 1][j] > dp[i][j - 1])
                           ? dp[i - 1][j]
                           : dp[i][j - 1];
            }
        }
    }

    printf("%d\n", dp[m][n]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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